Republic of the Philippines
UNIVERSITY OF SOUTHEASTERN PHILIPPINES
Iñigo St., Bo. Obrero, Davao City 8000
Telephone: (082) 227-8192
Website: www.usep.edu.ph
Email: president@usep.edu.ph

Form No.	FM-USeP-MID-01
Issue Status	01
Revision No.	00
Date Effective	05 January 2026
Approved by	President

COURSE SYLLABUS IN AEE 321L – MICROPROCESSOR SYSTEMS LAB

PART I. UNIVERSITY INFORMATION

VISION	Premier research university transforming communities in the ASEAN and beyond.
MISSION	USeP shall contribute to inclusive growth and sustainable development through continual innovation in proactive academic programs, impactful research for utilization, economically empowering community services, and sustainable resource management.
GOALS	● Internationalized University programs and services. ● Innovation and RDE-nurturing environment. ● Proactive, globally competitive, and service-oriented graduates. ● Transformed vulnerable and disadvantaged groups and communities. ● Mobilized and optimized resources for sustainable development.
EDUCATIONAL PHILOSOPHY	
Nature of Education in the University	The University of Southeastern Philippines is a state university that contributes to the development of human capital and to fostering innovations to transform communities in Mindanao, the Filipino nation, and the world. As a community of scholars, we aim to develop life-long learners who, guided by conscience, actively engage in the pursuit of generating and extending knowledge through research, extension, and responsible professional practice.
Principles for Teaching and Learning in the University	In all our educational endeavors, we strive to be continually guided by the following principles: ● Teaching and learning is based on science and evidence stemming from experience and reflections on the teaching practice. ● A conducive learning environment is founded on integrity and mutual respect, where learning is a co-construction among scholars. ● The learner is an active participant in the learning process to develop oneself and others' potential. ● A transformative learning experience is grounded in critical and reflective thinking and provides opportunities for practice, improvement, and action.
INSTITUTIONAL GRADUATE ATTRIBUTES	
IGA1: LEADERSHIP SKILLS	Creates and inspires positive changes in the organization; exercises responsibility with integrity and accountability in the practice of one's profession or vocation.
IGA2: CRITICAL AND ANALYTICAL THINKING SKILLS	Demonstrates creativity, innovativeness, and intellectual curiosity in optimizing available resources to develop new knowledge, methods, processes, systems, and value-added technologies.
IGA3: SERVICE ORIENTED	Demonstrates concern for others, practices professional ethics, honesty, and exemplifies socio-cultural, environmental concern, and sustainability.
IGA4: LIFELONG LEARNING	Demonstrates enthusiasm and passion for continuous personal and professional development.
IGA5: PROFESSIONAL COMPETENCE	Demonstrates proficiency and flexibility in the area of specialization and in conveying information in accordance with global standards

PART II. PROGRAM INFORMATION

Program Name: Bachelor of Science in Electrical Engineering

PROGRAM GRADUATE OUTCOMES			
PGO	**Description**	**Performance Indicator**	**IGA Aligned to**
PGO b	Design and Conduct experiments, as well as to analyze and interpret data	b.1. Perform laboratory experiments as prescribed by standard procedures	IGA2, IGA5
PGO c	Design a system, component, or process to meet desired needs within realistic constraints, in accordance with standards	c.1. Design agricultural structures and electrical systemms using computer-aided design software.	IGA2, IGA5
PGO g	Communicate effectively	g.2. Articulate the convention and style in written communication that conveys complete thought and information	IGA5
PGO k	Use techniques, skills, and modern engineering tools necessary for engineering practice.	k.3. Use modern technology in performing engineering practice.	IGA4

PART III. DOCUMENT INFORMATION

Version Number	:	2026–1.0
Version Note	:	✓ New ☐ Minor Revision ☐ Major Revision ☐ No Revision
Describe Revision *(if applicable)*	:	
Semester	:	Second Semester
Academic Year	:	2025-2026

PART IV. FACULTY INFORMATION

Faculty Name	:	ELWIN JAY J. TANO
Email Address	:	ejjtano0120241110@usep.edu.ph
Contact Number	:	09262995868
Office	:	College of Engineering
Consultation Hours	:	Monday-Friday, 8:00 AM – 05:00 PM (excluding lunch break)

PART V. COURSE INFORMATION

Course Code	:	AEE 311L
Course Title	:	Microprocessor Systems Lab
Pre-requisite	:	AEE 311 – Microprocessor System (co-requisite)
Credit	:	1.0 Unit – Laboratory
Level	:	Third Year

SDG Alignment:

SDG	Description of Alignment
9: Industry, Innovation, and Infrastructure	The course develops foundational competencies in microprocessor architecture, instruction execution, and I/O interfacing, enabling students to design and troubleshoot embedded systems used in industrial automation, control systems, and digital infrastructure.
11: Sustainable Cities and Communities	To design and interface microcontroller-based systems involving sensors, timing, and control, the course supports applications in smart systems such as automated lighting, monitoring, and basic control technologies that contribute to efficient and sustainable community infrastructure.

Course Description:

The course is about the laboratory component of Microprocessor Systems. It includes history and evolution, principles, and applications of microprocessors. The focus is on the basic understanding of the architectural design, functional parts, operations, function, and programming. It also covers the study of various types of microprocessors and the fundamental concepts of microprocessors.

Workload of Students:

With **18 weeks** at **3 contact hours** per week (**54 contact hours total**), the expected student workload is as follows:

- Complete **10 laboratory activities** with **post-laboratory oral recitations**
- **Weekly progress reports** for concept and functional project designs.
- **Practical Examination**
- **Concept project design**
- **Functional project design**

In instances where face-to-face sessions are impractical, simulated experiments or their equivalent will be provided.

Course Learning Outcomes (CLOs)

On the completion of the course, the student is expected to be able to do the following:

Code	Course Learning Outcomes	PGO Aligned to
CLO1	Students will be able to conduct hands-on activities to build working prototypes, troubleshoot, interpret collected data, and validate the functionality of Arduino, ESP32, and/or any microcontroller system within specified parameters.	PGO b
CLO2	Students will be able to create technical systems with specific requirements while considering factors such as budget, environment, and user impact, ensuring the systems perform as expected within defined limitations.	PGO c
CLO3	Students will be able to compile clear and organized reports by following appropriate formats for documenting projects and results, ensuring their reports are logically structured and communicate findings completely.	PGO g
CLO4	Students will be able to apply contemporary tools and methods in electronic design when constructing and testing various components and systems for communication, demonstrating proficiency in their use.	PGO k

PART VI. ASSESSMENT ACTIVITIES

Formative Assessments:

Code	Assessment Activity	Description and Other Details	CLO Aligned to
FA1	Progress Report	The progress report evaluates students' ongoing development of their IoT project, focusing on conceptual clarity, system design evaluation, implementation status, and documentation quality. Feedback from this assessment guides improvements before final project completion.	CLO2, CLO3
FA2	Oral Recitation	Oral recitation assesses students' ability to verbally explain concepts, justify design decisions, and respond to technical questions related to laboratory activities and project development. It promotes active engagement and immediate feedback.	CLO1, CLO2, CLO3, CLO4

Summative Assessments:

Code	Assessment Activity	Description and Other Details	CLO Aligned to
SA1	Laboratory Report	The laboratory report documents the objectives, procedures, observations, results, and analysis of each laboratory activity, emphasizing systematic experimentation and clear technical communication. Through hands-on work with microcontrollers, sensors, and communication technologies, the laboratory activities support SDG 9 by developing foundational skills in innovation and technological infrastructure, while also contributing to SDG 11 by enabling data-driven and technology-assisted solutions relevant to smart, safe, and efficient communities.	CLO1, CLO3, CLO4
SA2	Practical Exam	Evaluates students' ability to design, implement, and troubleshoot a functional microcontroller-based system using appropriate engineering tools under time-constrained conditions.	CLO3, CLO4
SA3	Concept Project Design	The concept project design presents the proposed IoT-based solution that addresses a real-world problem through innovative use of technology. It supports SDG 9 by fostering creativity, system thinking, and early-stage engineering innovation, and aligns SDG 11 by encouraging solutions that improve community functionality, safety, sustainability, or quality of life within urban and local environments.	CLO2, CLO3

SA4	Functional Project Design	The functional project design details the translation of an approved concept into a working IoT system, highlighting hardware-software integration, system reliability, and operational readiness. This assessment advances SDG 9 by strengthening students' capabilities in building and deploying functional technological systems and reinforces SDG 11 by focusing on practical applications that support smart infrastructure, efficient resource use, and responsive community systems.	CLO1, CLO2, CLO3, CLO4

PART VII. MEASUREMENT SYSTEM

Assessment Activity: FA1: Progress Report

Area to Assess	Excellent (5)	Very Good (4)	Satisfactory (3)	Needs Improvement (2)	Unsatisfactory (1)	Unacceptable (0)
Project Concept Clarity & Objectives (CLO2)	Project goals are clear, feasible, and well-justified; strong alignment with identified needs.	Goals are clear and mostly justified.	Goals are understandable but lack refinement.	Goals are vague or incomplete.	Goals are unclear or inappropriate.	No concept presented.
System Design Progress (Hardware & Software) (CLO3)	Design is clearly updated, coherent, and technically sound with diagrams.	Design updates are mostly complete with minor gaps.	Adequate design progress but lacking detail.	Minimal or poorly explained design progress.	Design is incorrect or inconsistent.	No design shown.
Implementation Status & Evidence (CLO3)	Clear evidence of implementation with testing results and validation.	Implementation progressing with partial testing.	Implementation is evident but weakly validated.	Limited implementation with little evidence.	Minimal or incorrect implementation.	No implementation.
Technical Documentation & Written Communication (CLO3)	Report is well-organized, concise, and technically accurate.	Well-written with minor issues.	Understandable but needs better organization.	Poor clarity and structure.	Very weak communication.	No report submitted.

Assessment Activity: FA2: Oral Recitation

Area to Assess	Excellent (5)	Very Good (4)	Satisfactory (3)	Needs Improvement (2)	Unsatisfactory (1)	Unacceptable (0)
Conceptual Understanding of IoT and Microcontroller Topics (CLO1)	Demonstrate deep understanding and accurate explanations.	Shows clear understanding with minor errors.	Basic understanding with some inaccuracies.	Limited understanding.	Major misconceptions.	No answer.
Explanation of Design Choices and Logic (CLO2)	Clearly explains and justifies technical decisions.	Mostly clear explanations.	Adequate but weak justification.	Poor explanation.	Unable to justify decisions.	No explanation.
Use of Technical Terminology and Clarity of Expression (CLO3)	Uses correct terminology with clear, logical delivery.	Minor issues in terminology or clarity.	Understandable but inconsistent.	Unclear or poorly structured responses.	Very unclear communication	No response
Responsiveness to Questions and Follow-Ups (CLO4)	Respond accurately and confidently without assistance.	Responds correctly with minimal prompting.	Answers with some guidance.	Struggles to respond appropriately.	Unable to respond.	No attempt.

Assessment Activity: SA1: Laboratory Report

Area to Assess	Excellent (5)	Very Good (4)	Satisfactory (3)	Needs Improvement (2)	Unsatisfactory (1)	Unacceptable (0)
Adherence to Procedure (CLO1)	Follows the experimental procedure with no deviation or error, adhering strictly to the steps and safety protocols.	Follows the procedure correctly with minor deviations or errors that do not impact the outcome.	Follows most of the procedure but makes notable mistakes that could affect the experiments' outcome.	Frequently deviates from the procedure or misses several steps; safety concerns are noted.	Significant deviations from the procedure; many steps are omitted or done incorrectly.	Does not attempt the procedure or deviate extensively without following critical safety measures.
Document and report experimental results clearly and professionally (CLO3)	The report is organized and written clearly, presenting results logically, with precise explanations and appropriate data visuals.	The report is mostly well-organized, with clear writing and few omissions or errors.	The report is understandable but may lack clarity in sections or omit some relevant details or explanations.	The report is disorganized; missing important data or explanations, making it hard to follow.	The report is poorly written, hard to understand, or lacks critical elements necessary for interpretation of results.	No report submitted or report does not contain any meaningful information.
Problem Identification and Resolution (CLO4)	Independently identifies issues during the experiment and resolves them effectively, ensuring correct results.	Identifies and resolves issues with minimal assistance, though may need some support in complex situations.	Identifies issues but requires guidance or time to resolve; partial success in resolving problems.	Struggles to identify problems or fails to address them effectively, requiring significant assistance.	Cannot identify or solve the key problems in the experiment; heavily reliant others for solutions.	Does not attempt to troubleshoot or resolve issues.
Timeliness and meeting deadlines (CLO3)	No delay, submitted on time as required.	Submitted on time with a delay of 1 day or less or communicated minor delay promptly.	Missed deadline by 2-4 days; informed instructor late.	Missed deadline by 5-6 days; minimal communication.	Missed deadline of 7-10 days; no prior communication regarding delay.	No submission or submission over 10 days late.

Assessment Activity: SA2: Practical Examination

Area to Assess	Excellent (5)	Very Good (4)	Satisfactory (3)	Needs Improvement (2)	Unsatisfactory (1)	Unacceptable (0)
System Implementation & Integration (CLO3)	System is fully functional, correctly integrated, and meets all tasks requirements without errors.	System works with minor issues that do not affect functionality.	System functions but require adjustments or corrections.	System partially works; several functions fail.	System rarely works or is incorrectly assembled.	No system implemented.
Circuit Assembly & Hardware Configuration (CLO3)	Circuit is correctly wired, safe, neat, and stable on first attempt.	Minor (wiring) errors but corrected quickly.	Circuit work but it shows poor organization or minor safety issues.	Multiple wiring errors require guidance.	Major safety or safety issues.	No circuit assembled.
Code Logic & Execution (CLO3)	Code is correct, efficient, well-structured, and executes flawlessly.	Code works with minor logical or structural issues.	Code executes but with noticeable inefficiencies or bugs.	Code partially works with significant errors.	Code does not function.	No code submitted.
Troubleshooting & Use of Engineering Tools (CLO4)	Independently identifies and resolves issues using appropriate tools and methods.	Resolves issues with minimal guidance.	Can troubleshoot with moderate assistance.	Struggles to identify issues without help.	Unable to troubleshoot effectively.	No troubleshooting attempt.

Assessment Activity: SA3: Project Proposal Design

Area to Assess	Excellent (5)	Very Good (4)	Satisfactory (3)	Needs Improvement (2)	Unsatisfactory (1)	Unacceptable (0)
Problem Definition and Objectives (CLO2)	Comprehensive problem definition addressing key constraint.	Clear problem definition with limited depth.	Well-defined but lacks context or constraints.	Basic problem definition, missing details.	Unclear, incomplete, or poorly defined problem.	No clear (problem) statement.
Solution Approach and Feasibility (CLO2)	Well-defined, feasible solution that integrates key constraints.	Plausible solution considering key constraints.	Viable solution with some feasibility concerns.	Basic solution but lacks critical feasibility.	Vague or unrealistic solution, lacking feasibility.	No approach or solution proposed.
Literature Review and Related Work (CLO3)	Thorough and insightful review, critically engaging with related work.	Detailed review, some critical analysis needed.	Good review with some analysis, but lacking depth.	Basic, limited review, missing analysis.	Minimal or irrelevant review lacks context.	No related work cited.
Technical Plan and Timeline (CLO2)	Detailed, well-structured plan and realistic timeline.	Clear plan with a realistic timeline, few gaps.	Well-organized plan, missing task-specific details.	Some planning with unclear task breakdown.	Unclear plan, unrealistic or non-detailed timeline.	No plan or timeline provided.
Ethical and Sustainability Considerations (CLO2)	Thorough ethical and sustainability consideration integrated throughout the design.	Strong but not fully detailed ethical and sustainability considerations.	Ethics and sustainability considered, but some gaps remain.	Limited consideration of ethics/sustainability.	Brief mention, poorly applied to the design.	No ethical or sustainability considerations.

Assessment Activity: SA4: Functional Project Design

Area to Assess	Excellent (5)	Very Good (4)	Satisfactory (3)	Needs Improvement (2)	Unsatisfactory (1)	Unacceptable (0)
Clarity and Quality of Presentation (CLO3)	Excellent clarity, flow, and impactful visuals.	Good clarity and flow, with some impact from visuals.	Clear, but lacks smooth transitions or full engagement.	Key ideas presented but lack clarity and flow.	Struggles to present clear ideas, poor visuals.	Presentation lacks clarity, structure, and focus.
Data Analysis and Interpretation (CLO1)	Comprehensive analysis and deep insights fully support conclusions.	Good analysis, but some insights or conclusions lacking.	Accurate analysis but needs more interpretation or details.	Some data analysis lacks depth or clarity.	Limited, inaccurate, or superficial analysis.	No analysis or interpretation provided.
Technical Execution and Problem-solving (CLO4)	Excellent proactive problem-solving and advanced engineering solution.	Strong problem-solving with efficient technical solutions.	Clear problem-solving with minor issues unresolved.	Issues acknowledged and solved but lacks depth.	Basic problems recognized but not solved effectively.	No problem solving demonstrated.
Impact and Outcomes Presentation (CLO2)	Detailed and well-documented outcomes linked directly to the objectives.	Solid results with strong connection to objectives.	Clear outcomes are linked to project objectives.	Basic outcomes presented, connection to objectives weak.	Vague or unclear discussion of outcomes.	No outcomes presented.
Professional and Ethical Considered Considerations (CLO2)	Exemplary professionalism with full ethical considerations.	Demonstrates professionalism with some ethical discussion.	Discusses ethics in the project but lacks depth.	Briefly acknowledged ethics, limited application.	Poor addressed ethical considerations.	No attention to ethics or professionalism.

PART VIII. GRADING SYSTEM AND ASSESSMENT PLAN:

The final grade in this course will be composed of the following items and their weights in the final grade computation:

Assessment Item	Percentage of Final Grade	CLO1	CLO2	CLO3	CLO4
SA1	15%	Laboratory Report Rubric		Laboratory Report Rubric	Laboratory Report Rubric
SA2	25%			Practical Exam Rubric	Practical Exam Rubric
SA3	25%		Concept Design Rubric	Concept Design Rubric	
SA4	35%	Functional Project Rubric	Functional Project Rubric	Functional Project Rubric	Functional Project Rubric
TOTAL	**100%**				

Transmutation	:	Base-50
Passing Grade	:	3.00
Passing Grade Conditions	:	Students shall submit all progress reports, laboratory reports, take the practical exam, present concept design and functional project along with the prototype, and must have an attendance of not less than 80% in the class. In case of lacking output or examination, a grade of INC (Incomplete) shall be given in accordance with the existing rules and regulations reflected in the student handbook. In case of incurring absences of more than 20% of the total expected number of attendances, a grade of UW (Unauthorized Withdrawal) will be reflected in the grade sheet.

Grade Equivalence Table:

< 75 = 5.00	75-76 = 3.00	77-79 = 2.75	80-82 = 2.50	83-85 = 2.25
86-88 = 2.00	89-91 = 1.75	92-94 = 1.50	95-97 = 1.25	98-100 = 1.00

PART IX. LEARNING PLAN

In order to achieve the outcomes of this course, learners will go through this learning plan:

Intended Learning Outcomes (ILO)	Course Learning Outcome (CLO)	Time Frame	Topics	Teaching-Learning Activities (TLA)	Required Reading	Assessment Activities
• Recognize the university VMGO & Institutional Graduate Attributes. • Appreciate the significance of PEOs and PGOs of the programs. • Recognize the rules and regulations of the course and the grading system.		1 hour	• Course Orientation • VMGO & IGA of the University • PEOs & PGOs of the program. • Rules and regulations of the course & grading system • Presentation of the course outline	• On-site or online meeting	• UseP Student Handbook (2016 edition)	• N/A
• Understand the basic functionality and setup process of Arduino hardware and the Arduino IDE. (CDIO Skills: 2.1, 2.3, 1.2)	CLO1 CLO3 CLO4	3 hours	• **LABORATORY ACTIVITY 1: Introduction to Arduino and IDE Setup**	• Interactive demonstration and discussions. • Hands-on Laboratory Sessions • Collaborative Laboratory Work • Oral Recitation	• Laboratory Manual • Ref04-06	• FA2 – Oral Recitation • SA1 – Laboratory Report
• Develop the ability to interface and program input devices, such as buttons and sensors, and analyze their behavior. (CDIO Skills: 2.1, 4.4, 1.3)		3 hours	• **LABORATORY ACTIVITY 2: Using Digital and Analog Inputs**	• Interactive demonstration and discussions. • Hands-on Laboratory Sessions • Collaborative Laboratory Work • Oral Recitation	• Laboratory Manual • Ref04-06	• FA2 – Oral Recitation • SA1 – Laboratory Report
• Design and implement a feedback system using sensors to create a real-time response to environmental inputs. (CDIO Skills: 4.4, 2.1, 1.2)		6 hours	• **LABORATORY ACTIVITY 3: Sensor-based Feedback Loop with Arduino**	• Interactive demonstration and discussions. • Hands-on Laboratory Sessions • Collaborative Laboratory Work • Oral Recitation	• Laboratory Manual • Ref04-06	• FA2 – Oral Recitation • SA1 – Laboratory Report
• Learn to configure and program the ESP32 microcontroller for basic wireless control applications. (CDIO Skills: 2.3, 2.1, 4.4)		3 hours	• **LABORATORY ACTIVITY 4: Introduction to ESP32 and Wi-Fi Setup**	• Interactive demonstration and discussions. • Hands-on Laboratory Sessions • Collaborative Laboratory Work • Oral Recitation	• Laboratory Manual • Ref01, Ref04-06	• FA2 – Oral Recitation • SA1 – Laboratory Report
• Construct and analyze a wireless system for real-time data acquisition		6 hours	• **LABORATORY ACTIVITY 5: Wireless Data Monitoring with ESP32**	• Interactive demonstration and discussions.	• Laboratory Manual • Ref01, Ref04-06	• FA2 – Oral Recitation

Intended Learning Outcomes (ILO)	Course Learning Outcome (CLO)	Time Frame	Topics	Teaching-Learning Activities (TLA)	Required Reading	Assessment Activities
and visualization using the ESP32. (CDIO Skills: 4.4, 2.1, 1.3)			• **Optional: Discussion/Seminar with Faculty Intern/COIL**	• Hands-on Laboratory Sessions • Collaborative Laboratory Work • Oral Recitation		• SA1 – Laboratory Report
• Project Proposal Presentation (CDIO Skills: 3.2, 4.3, 3.1)	CLO2 CLO3	3 hours	**PROJECT PROPOSAL PRESENTATION**	• **Project Proposal Presentation**		• **FA1 – Progress Report** • **SA3 – Concept Project Design**
• Implement a cloud-based IoT system using ESP32 for data publishing and remote control. (CDIO Skills: 3.2, 4.3, 3.1)	CLO1 CLO3 CLO4	3 hours	• **LABORATORY ACTIVITY 6: Cloud-Connected IoT System**	• Interactive demonstration and discussions. • Hands-on Laboratory Sessions • Collaborative Laboratory Work • Oral Recitation	• Laboratory Manual • Ref01, Ref04-06	• FA2 – Oral Recitation • SA1 – Laboratory Report
• Design a safe and functional IoT system for home automation using ESP32 and relay modules (CDIO Skills: 4.4, 2.1, 2.3)		6 hours	• **LABORATORY ACTIVITY 7: Home Automation using Relays and ESP32**	• Interactive demonstration and discussions. • Hands-on Laboratory Sessions • Collaborative Laboratory Work • Oral Recitation	• Laboratory Manual • Ref01, Ref04-06	• FA2 – Oral Recitation • SA1 – Laboratory Report
• Integrate Arduino and ESP32 for seamless data exchange and develop a multi-device system for data visualization (CDIO Skills: 4.4, 2.3, 1.3)		3 hours	• **LABORATORY ACTIVITY 8: Arduino-ESP32 Communication for Data Integration**	• Interactive demonstration and discussions. • Hands-on Laboratory Sessions • Collaborative Laboratory Work • Oral Recitation	• Laboratory Manual • Ref01 -06	• FA2 – Oral Recitation • SA1 – Laboratory Report
• Combine RFID and ultrasonic sensors into a multi-functional system for access control and obstacle detection. (CDIO Skills: 4.4, 2.1, 1.2)		3 hours	• **LABORATORY ACTIVITY 9: Multi-technology Access and Obstacle Detection System**	• Interactive demonstration and discussions. • Hands-on Laboratory Sessions • Collaborative Laboratory Work • Oral Recitation	• Laboratory Manual • Ref01 -06	• FA2 – Oral Recitation • SA1 – Laboratory Report
• Implement time-sensitive tasks and data logging mechanism using RTC		6 hours	• **LABORATORY ACTIVITY 10: Time-based Automation and Data Logging**	• Interactive demonstration and discussions.	• Laboratory Manual • Ref01 -06	• FA2 – Oral Recitation

Intended Learning Outcomes (ILO)	Course Learning Outcome (CLO)	Time Frame	Topics	Teaching-Learning Activities (TLA)	Required Reading	Assessment Activities
modules and communication protocols. (CDIO Skills: 4.4, 2.1, 2.3)				• Hands-on Laboratory Sessions • Collaborative Laboratory Work • Oral Recitation		• SA1 – Laboratory Report
• Pass the practical examination (CDIO Skills: 4.4, 2.3, 1.2)	CLO3 CLO4	3 hours	**PRACTICAL EXAMINATION**	• **Practical Examination**		• **SA2 – Practical Examination**
• Pass the final examination. (CDIO Skills: 3.2, 4.6, 3.1)	CLO1 CLO2 CLO3 CLO4	3 hours	**FUNCTIONAL PROJECT DESIGN PRESENTATION**	• **Functional Project Design Presentation**		• **FA1 – Progress Report** • **SA4 – Functional Project Design**

PART X. REFERENCES/ARTICLES TO READ

Code	Reference/Article
Ref01	Lopez, S.C., Rivas, P., & Tabuclin, J.R. (2024). *IoT-Enhanced Intravenous (IV) Therapy Real-Time Management System: Drip Rate, Fluid Level, and Pulse Detection with NodeMCU-ESP8266 Microcontroller and Sensor Fusion* (Adviser: R.C. Ucat)
Ref02	Heruela, P.M. & Pedregosa, V.P. (2022). *IoT-Based Medicine Administration with Dispensing System* (Adviser: R.C. Ucat)
Ref03	Jovenir, P. et.al. (2016). *Programmable Illumination Control Device (PIC'D)* (Adivser: R.C. Ucat)
Ref04	Monk, S. (2016). *Programming Arduino™: Getting Started with Sketches (2nd ed.)*. McGraw-Hill Education
Ref05	Blum, J. (2013). *Exploring Arduino: Tools and Techniques for Engineering Wizardry*. John Wiley & Sons, Inc.
Ref06	Margolis, M. (2011). *Arduino Cookbook*. O'Reilly Media, Inc.

PART XI. CLASSROOM POLICIES

1. **Attendance**. Classroom attendance is an integral part of the educational experience; therefore, it is the student's responsibility to attend class. Attendance will be attended at every meeting. If the student missed any laboratory session with no valid excuse, it will be ineligible to have a laboratory activity grade on that session.
2. **Behavior and Decorum**. Respectful, inclusive, and professional behavior is always expected. The use of polite and non-discriminatory language is strictly observed in line with the University's commitment to a gender-fair and inclusive learning environment, and adherence to RA 11313, also known as the Safe Spaces Act.
3. **Laboratory Conduct and Adherence to Safety Protocols**. All equipment and materials must be handled with care; any misuse or negligence will result in reduction in the grade of laboratory activity. Students are expected to follow mandatory safety protocols, especially proper handling of electronic devices and checking the actual circuit before energizing.
4. **Submission of Coursework**. Extensions and late submissions of requirements will only be considered if done in writing to the lecturer at least two days before the due date. Generally, no extension will be given for the assignments. Please contact your subject adviser if you encounter any problems.
   Note: (Submission / Resubmission):
   - *Should work be submitted after the due date without written approval for an extension, a penalty of 5% deduction from the grade will be incurred for every 2 days it is late.*
   - *Should the work be assessed as being close to a PASS grade, but the standard has not yet been fully met, the lecturer may offer the opportunity to resubmit.*
5. **Completing Progress Reports**. Students are expected to submit progress reports in relation to their midterm and final requirements to monitor progress. The students are expected to submit progress reports at the end of the week (Friday). Any missed submission will be given an INC grade.
6. **Use of AI Tools**. The responsible use of generative AI tools (e.g., ChatGPT, Grammarly, Quillbot) is permitted only when explicitly allowed by the instructor for specific tasks. All AI-assisted work must be properly cited or declared. Utilizing generative AI tools for generating code, especially during laboratory activities is strictly prohibited. Any group who will be caught will be given half of the points on that activity.
7. **Academic Honesty**. All USePians are bound to follow the rules and regulations stipulated in the student handbook of the University and the (College/School). All University policies regarding academic honesty apply to this course. Academic dishonesty includes, but is not limited to, cheating in any form, plagiarism, and similar offenses.
   - Cheating is a serious offense. If you cheat on any assignment or examination, the minimum penalty is a grade of zero for that work. Your conduct may also be referred to the Office of Student Affairs and Services (OSAS) for appropriate sanctions under university policy.
   - Students are encouraged to work together in exchanging ideas and discussing their assignments in general. However, it is also expected that the work of analyzing data and writing reports will be done in their respective groups. If reports or data have been copied from another student, both students will receive a mark of zero on that examination, test, report, or assignment.
   - Submitting AI-generated content as original work without disclosure is considered academic dishonesty. The acceptable level of matched content using university-accepted plagiarism detection software is 15%. Any paper exceeding this threshold may be subject to rejection or revision.
8. **Conduct of Quizzes and Examinations.** Every quiz and exam, students must keep the jacket of their calculators, smartphones, and other things irrelevant to the quiz or exam in their respective bags. If the student caught with any of those objects during the quiz and/or examination proper, it would serve as cheating and dealt accordingly in accordance with the policies imposed by the latter. Instructions must be observed with utmost compliance. Unable to do so will lead to consequences (e.g., deduction of points or answers will not be recognized). During the solution part, rounding up is not recommended as it will accumulate rounding-off errors to the final answer. Final answers must be in three decimal places. Math plate is observed to be used during quizzes and bluebook for term examinations. In case the former is unavailable, bluebook can be used as a substitute.
9. **Use of Calculators.** Allowed calculators must follow the Professional Regulatory Commission (PRC) recommendations, see https://www.prc.gov.ph/node/6984 for the full list of non-programmable calculators allowed by the Commission.
10. **Cleanliness and Energy Conservation**. After each laboratory session, ensure that everything is in order and all plugged devices and lights are turned off before leaving the classroom.

Prepared by:	Reviewed by:	Approved:
ELWIN JAY J. TANO Faculty	**CHARITO O. CLARO** Program Head – Electrical Engineering	**JOIE D. CATAYTAY** Dean